

Luminous City: A Manifestational Ontology of Smart City Planning and Construction

*An applied monograph on urban planning based on the Ontology of the
Luminous Will*



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Abstract

This book is an applied ontological treatise based on the Ontology of the Luminous Will. It discusses the fundamental basis, value boundaries, and methodological paths for smart city planning and construction. Its central concern is not whether smart cities can build more platforms, connect more data, or deploy more algorithms, but whether the differences of life within the city can be seen, borne, governed within admissible boundaries, and reorganized after structural exhaustion.

The Luminous City is not a visual spectacle created by lights, screens, exhibition halls, or digital twin models. In this book, radiance has a strict ontological meaning: it refers to the process by which differences move from an unmanifest state into a visible, perceptible, discussable, and bearable state. The real question of smart cities is therefore whether manifestation is real, whether bearing structures are adequate, whether regulation has boundaries, and whether renewal remains possible.

This book maintains a fundamental judgment: the city is not a machine, but a complex vessel through which differences are continuously manifested. Within the city exist roads, buildings, platforms, institutions, and algorithms, as well as bodies, memories, fears, care, desires, conflicts, ecological pressures, and public life. If a smart city focuses only on the former while obscuring the latter, it becomes an efficient system of concealment; only when the latter can enter structures of bearing can the city become a true Luminous City.

In this book, will does not refer to a psychological subject or unified consciousness, but to the structural tendency by which differences within a constraint system persist without reverting to an unmanifest state. Radiance does not refer to visual brightness, but to the manifestational process by which differences enter a visible, perceptible, discussable, and sustainable state. This terminological restriction governs the whole book.

Keywords:

Luminous City; Manifestational Ontology; Smart City; Urban Planning; Urban Governance; Urban Radiance; Urban Topology; Digital Twins; Urban Resilience; Human-Centered Cities; Algorithmic Governance; Radiant Ethics.

Preface

Why This Book Was Written

This book was written from a practical unease: many smart city projects become increasingly sophisticated as technological systems, yet the lives they are meant to serve remain only partially visible. Platforms can be integrated, screens can be enlarged, models can be refined, and indicators can be improved, while old residential compounds, bodily exhaustion, inaccessible services, governance fatigue, and fragile public trust continue to exist in the shadow of technical success.

The problem is not that technology has entered the city. The problem is that technology is too often treated as if it were the city itself. This book therefore asks how smart city construction can return from platform-centered engineering to the manifestation and bearing of urban life.

Theoretical Position

The conceptual framework of this book derives from the author's Ontology of the Luminous Will. In the urban context, will does not mean a collective psyche or a unified municipal subject; it means the persistence of differences that refuse to disappear into silence. Radiance does not mean illumination, display, or spectacle; it means that these differences become visible, discussable, answerable, and capable of being borne.

Accordingly, the book reads the smart city through a sequence of unmanifest difference, persistent urban will, radiant manifestation, urban vessels, topology and admissibility, exhaustion and concealment, ethical constraint, and reorganization. This sequence is meant to help planners, administrators, researchers, and citizens judge whether a smart city project truly increases the city's capacity to see, bear, protect, and renew life.

How to Read This Book

The book does not provide a single engineering scheme for any particular city and does not replace statutory planning, feasibility study, design review, or legal compliance. It provides an ontological and methodological calibration framework. Readers may use it as a conceptual map for diagnosing blind spots, reviewing smart city projects, designing governance mechanisms, and asking whether the visible parts of a system are actually connected to structures of bearing.

The Introduction that follows presents the full analytic architecture. The chapters then move from ontological grounding to spatial, data, algorithmic, institutional, and community vessels; from admissible boundaries to exhaustion and alienation; and finally to ethics, health checks, digital twins, operations, renewal, evaluation, and continuous reorganization.

Introduction

A Manifestational Ontology of the Smart City

The Ontological Problem of the Smart City

Contemporary smart city construction is often marked by information and communication technologies, the Internet of Things, artificial intelligence, cloud computing, digital twins, urban operation centers, and platform-based governance. These technologies do improve urban sensing, dispatching, and response capacities, and they also move urban governance from experience-based management toward data-assisted decision-making. Definitions of smart sustainable cities offered by international organizations commonly emphasize the use of ICT and other means to improve quality of life, urban operational efficiency, and public services, while also considering the economic, social, environmental, and cultural needs of present and future generations.

Yet technical definitions do not exhaust the ontological meaning of the smart city. A city may possess an advanced sensing network and still fail to see the difficulty of an elderly person going downstairs. It may possess a real-time traffic screen and still fail to change the daily bodily exhaustion of commuters. It may possess public security algorithms and yet invisibly amplify bias and excessive surveillance. It may have impressive indicator rankings and still allow real life to be concealed by forms, tables, and the logic of project acceptance.

For this reason, the smart city needs not only engineering, management science, and information science, but also ontology. It must ask: how does the city manifest? What kinds of difference are recognized? What kinds of structure can bear them? What boundaries must be respected? What kinds of consumption and exhaustion are occurring? What kinds of renewal can allow the city to regain vitality?

The Theoretical Axis of This Book

This book unfolds along the path of the unmanifest precondition, the persistence of difference, radiant manifestation, the urban vessel, topology and admissibility, exhaustion and concealment, ethical constraint, and reorganization and renewal. This path is not a conceptual decoration added from outside to the smart city. It is a structural description of the process by which the city exists. The city is not first a completed object and only

then processed by technology; the city becomes a city through the continuous manifestation of multiple differences.

The unmanifest is not nonexistence. It is merely that which has not yet entered visible structures, discursive structures, indicator structures, or institutional structures. Children's activity spaces, mobility barriers for disabled persons, women's sense of safety at night, the vulnerability of underground pipe networks, heat island effects, fatigue in grassroots governance, and fractures in public services for low-income residents are all real urban differences that may remain unmanifest for a long time.

Radiance means that these differences begin to appear. But appearance alone is not enough. Without space, facilities, institutions, finance, algorithmic audit, public participation, and community support, the fact that difference is seen may become only a painful exhibition. The core of the smart city is therefore not sensing capacity, but bearing capacity.

The Boundary of This Book

This book does not provide a concrete construction scheme for any particular city, nor does it replace statutory planning, engineering design, or project feasibility studies. It provides a theoretical framework that can be used to judge, organize, and correct smart city construction. Concrete planning must still be combined with city size, fiscal capacity, industrial structure, population structure, ecological conditions, governance systems, and legal norms.

Nor is this book anti-technology. On the contrary, it recognizes digital technology as an important vessel of contemporary urban manifestation. The question is not whether technology should enter the city, but in what capacity it enters: as a vessel that bears difference, or as a power machine that devours difference; as an aid to public judgment, or as a black box that escapes responsibility; as a means by which more people are seen, or as a means by which certain people are more efficiently concealed.

The final goal of this book is to reinterpret the smart city from a technological city into a Luminous City. A city is smart not because it knows everything and controls everything, but because it can allow real differences of life to continue shining within structures that can bear, negotiate, and renew them.

Part I

Ontological Grounding: The City Is Not a Machine, but a Field of Difference

Chapter 1

The Ontological Misunderstandings of the Smart City

The Misunderstanding of Techno-Centrism

The smart city is often described as an integration of technological systems: sensors collect, networks transmit, platforms aggregate, algorithms judge, screens display, and departments handle. At the engineering level, this description is necessary. At the ontological level, however, it easily produces a misjudgment: as if the city were merely an object waiting to be taken over by technology, and as if the denser the technology, the smarter the city.

Within the Ontology of the Luminous Will, technology is not the being of the city, but a vessel within the process of urban manifestation. The value of a vessel does not lie in its own complexity, but in whether it can bear difference. A platform that cannot respond to real needs is only a device of self-display. An algorithm that cannot accept public scrutiny is only a black box of responsibility transfer. A data center that cannot enter planning and service is only illuminated silence.

The Misunderstanding of Data-Centrism

Data can manifest urban conditions, but data is not the city itself. Traffic volume, energy consumption, passenger flow, police reports, complaints, temperature, waterlogging points, and service radius can all become partial expressions of urban radiance. Yet data always carries boundaries of collection, modes of classification, model assumptions, and institutional preferences. The most vulnerable people in a city often leave the fewest computable traces; the most important feelings in a city may not be directly captured by indicators.

If the smart city treats data as the only reality, it excludes from the city forms of life that cannot be expressed by data. The Luminous City must recognize the value of data manifestation, but it must also recognize the irreplaceability of experiential manifestation, spatial manifestation, public-issue manifestation, and bodily manifestation.

The Misunderstanding of Control-Centrism

Many smart city projects understand intelligence as an increase in control capacity: controlling traffic, risk, population movement, community order, and emergencies. Such

an understanding turns the urban operation center into an urban control room and turns residents from members of an urban community into objects to be dispatched, predicted, and managed.

But the city is not a militarized system, nor is it a single objective function. The city possesses openness, diversity, contingency, and publicness. Intelligence is not the elimination of uncertainty, but the maintenance of bearing capacity within uncertainty. The true wisdom of the city is regulation rather than domination, coordination rather than flattening, and early warning rather than total control.

The Misunderstanding of Indicator-Centrism

Evaluation standards and indicator systems for smart cities have necessary significance for construction, acceptance, and comparison. ISO 37122:2019 provides definitions and methods for smart city indicators, and GB/T 33356-2022 also offers an evaluation framework for new-type smart cities. Yet any indicator can project only part of urban radiance. The clearer an indicator becomes, the more it may conceal forms of life that are difficult to quantify.

Indicators should be understood as tools of manifestation, not as value itself. If a city builds for scores, connects data for acceptance, and selectively presents itself for rankings, it has already slipped from a Luminous City into an indicator city. Indicators must be constrained by an anti-indicator consciousness: wherever something is measured, one must ask who has not been measured; wherever something is ranked, one must ask what has been sacrificed behind the ranking.

Chapter Summary

The core of this chapter is to free the smart city from one-dimensional understandings based on technology, data, control, and indicators. The smart city certainly needs technology, but technology is only a vessel. It certainly needs data, but data is only a partial projection of radiance. It certainly needs evaluation, but evaluation cannot replace urban value.

This chapter therefore establishes the critical point of departure for the whole book: every scheme that understands the city as an object completely capturable by platforms, every scheme that understands intelligence as stronger control, and every scheme that understands indicators as value itself must be re-examined through manifestational ontology.

Implications for Planning and Construction

- Before a smart city project is initiated, an ontological review should be completed: what difference does the project seek to manifest, rather than what technology should first be procured?
- Evaluation systems should retain anti-indicator mechanisms to prevent real life from being sacrificed for rankings and acceptance procedures.
- Urban operation centers should be positioned as regulation centers rather than centers of total control.

Chapter 2

The City as a Gathering Field of Unmanifest Difference

The Unmanifest Is Not Nonexistence

The unmanifest does not mean nonexistence. It means only that something has not yet entered a state of being visible, sayable, computable, and answerable. In the city, many real problems remain unmanifest for a long time: old residential compounds without elevators, children without safe outdoor play space, night routes that make people afraid, underground pipe networks whose vulnerability appears during storms, and heat islands that place elderly people at higher risk in summer.

These problems do not appear suddenly. They have existed for a long time but have not been recognized as objects of planning. If the smart city collects only data that has already been defined by systems, it will find it hard to touch these silent structures. The first task of the Luminous City is to allow unmanifest urban wounds to enter public visibility.

Types of Urban Difference

Urban difference is not one-dimensional but multidimensional and interwoven. Spatial difference appears as center and periphery, accessibility and isolation, density and dispersion. Social difference appears in income, age, occupation, gender, bodily capacity, and digital capacity. Ecological difference appears in the uneven distribution of heat islands, waterlogging, air quality, and access to green space. Institutional difference appears in different levels of access to resources, services, and governance response.

The smart city cannot treat these differences as noise. Difference is the precondition of urban manifestation. Without difference, the city is only a homogeneous plane. With difference, the city has structure, direction, pressure, and possibility. The task of planning is not to eliminate all differences, but to judge which differences need protection, which need regulation, and which are producing exhaustion.

Silent Groups and Low-Data Existence

Many people live on the margins of smart systems. They do not fail to exist; rather, they exist in a low-data form. Elderly people may not use smartphones. Children may not have stable digital identities. Failed journeys by disabled persons are rarely recorded. The

actual radius of daily life for migrant workers often does not enter official platforms. Fractures in public services for low-income groups are often covered over by individual endurance.

If a city believes only platform data, these people are concealed twice: first by real structures, and then by the smart system. The Luminous City must actively search for low-data existence. Through community interviews, field surveys, participatory mapping, hotline review, grassroots observation, and public deliberation, silent groups must obtain pathways of manifestation.

The Starting Point of Planning

Traditional planning often begins from land use, transportation, population, industry, and facility allocation. These are certainly important. But the Luminous City requires planning first to ask: who has not been seen? What suffering has not entered the agenda? Which risks are treated as normal loss? Which lives have long been suppressed by the urban structure?

For this reason, smart city construction should not begin with platform procurement, but with the discovery of unmanifest difference. It should not first build an urban brain, but first ask whether many bodies in the city still cannot be heard. It should not first pursue high-precision models, but first confirm whether real life outside the model has channels into institutions.

Chapter Summary

This chapter shows that the deepest urban problems are often not problems already seen, but differences that have long not been recognized as problems. The unmanifest is not nonexistence; it is that which has not yet entered a state of being sayable, visible, and answerable.

If the smart city relies only on existing data, it easily repeats existing concealments. A true Luminous City must actively search for low-data existence, silent groups, and institutional blind spots, shifting the starting point of urban planning from the blueprint to unmanifest difference.

Implications for Planning and Construction

- Before smart city construction begins, a list of unmanifest differences should be organized, including low-data groups, hidden risks, and service fractures.

- Data analysis must be combined with community interviews, field surveys, and participatory mapping.
- Planning goals should shift from covering more systems to seeing more real life.

Chapter 3

Urban Will: A Non-Psychological Tendency of Persistence

The City Has No Personal Will

This book uses the term urban will not in order to personify the city, nor to say that the city possesses consciousness, desire, or a psychological center like a person. The city has no unified mind, no self capable of representing everyone, and no naturally correct overall purpose. Urban will is a structural concept: it refers to the tendency by which certain differences in the city continue without falling back into silence.

This restriction is crucial. If urban will is understood as a collective personality, it can easily provide rhetoric for centralized power and technological rule. If it is understood as the persistence of difference, it allows us to see the multiple voices, conflicts, and negotiations inside the city. At the urban level, the Luminous Will is not a great subject but a set of differential tensions that continually demand manifestation.

Will as the Persistence of Difference

Commuters continually search for shorter routes. Residents continually strive for public space. Elderly people continually demand age-friendly facilities. Industries continually gather around certain nodes. Memory continually resists demolition. Floods continually reveal the boundaries of drainage systems. Heat islands continually remind us of insufficient ecological bearing. All these persistent phenomena can be understood as will-like expressions of urban difference.

They are not psychological wills but structural tendencies. Will means that difference is unwilling to return to an unmanifest state and continues to express itself through congestion, complaints, migration, gathering, decline, disease, risk, or public issues. The smart city must learn to read these expressions rather than press them back into silence through stronger control.

The Conflict of Multiple Wills

Urban will is not singular but plural. Commerce wants higher footfall; residents want less noise. Vehicles want to pass more quickly; pedestrians want to stay more safely. Governance systems want visibility and control; individuals want privacy. Development

wants intensive use; ecology requires openness and buffering. The city is the tension-structure among these persistences of difference.

If the smart city seeks a single optimum, it sacrifices the plural nature of the city. The Luminous City does not aim to eliminate conflict. It aims to establish structures that can negotiate, regulate, and bear conflict. It recognizes real contradictions among different wills and uses institutions, space, and technology to prevent contradictions from collapsing into violence or concealment.

Planning as Translation of Will

Planning does not create urban will out of nothing. It translates the persistences of difference that already exist in the city. Transport planning translates the will of movement. Community planning translates the will of care. Ecological planning translates environmental boundaries. Historic preservation translates the will of memory. Public-service planning translates demands for fairness. Data governance translates the tension between visibility and privacy.

Good smart city planning does not make the city obey an abstract indicator. It allows multiple differences to find forms that can be expressed, judged, and borne. Planning therefore becomes the medium through which the Luminous Will enters the urban vessel.

Chapter Summary

This chapter restricts the concept of urban will. The city has no personality, consciousness, or unified psychological center. Urban will means only the tendency by which difference within urban structures continues without falling back into silence.

This restriction allows the Luminous City to avoid holistic and anthropomorphic misreadings. It treats the city as a field of conflict and negotiation among multiple persistences of difference, not as a single subject.

Implications for Planning and Construction

- Planning analysis should identify multiple urban wills rather than assume a single public goal.
- Commuting, complaints, migration, spatial contention, and memory preservation can all be treated as signals of persistent difference.

- Smart platforms should bear polyphonic expression and avoid compressing complex demands into a single work order.

Chapter 4

Urban Radiance: From Concealment to Visibility

Radiance Is Not Landscape Aesthetics

The Luminous City is easily misunderstood as a city of lights, night views, screens, or exhibition halls. In fact, radiance in this book does not refer to visual brightness. An urban screen can be bright and yet truly manifest nothing. A nightscape can be brilliant and yet conceal the fatigue of lower-level life. A digital model can be exquisite and yet remain disconnected from residents' experience.

The ontological meaning of radiance is that difference enters a state that is visible, perceptible, discussable, and capable of being borne. A seen drainage risk, a repaired accessible route, a reactivated community square, or a public-service mechanism that truly responds to residents' needs is closer to radiance than mere lighting.

Data Radiance

Data radiance is the most conspicuous mode of manifestation in the smart city. Traffic flows display road pressure. Energy curves display energy load. Heat maps display crowd gathering. Sensors display environmental change. Risk models display disaster possibility. Government data reflect service demand. Data gives visible form to many urban conditions that were previously scattered.

Yet data radiance must remain humble. It is one form of radiance, not all of it. Data may tell us that complaints in a district are low, but it cannot prove that there is no suffering there. It may show that the efficiency of a road has improved, but it cannot prove that pedestrians have more dignity. It may prove that the number of platform connections has increased, but it cannot prove that the city listens better.

Spatial Radiance

Urban radiance also manifests through space. Streets make movement visible. Squares make public life visible. Parks make rest and ecology visible. Schools and hospitals make service bearing visible. Community centers make care relations visible. Historic districts make memory visible. Space is not the background of data; it is an important medium of urban existence.

If smart city construction invests only in back-end platforms while neglecting streets, communities, parks, slow-traffic systems, and public-service spaces, it produces the paradox of a highly intelligent back end and an impoverished front stage. The Luminous City must bring technological radiance back to spatial radiance, so that people in the city can feel bearing in actual space.

Experiential Radiance

A sense of safety, belonging, dignity, place, neighborhood trust, cultural memory, and the capacity for public participation together constitute the experiential radiance of the city. These elements cannot always be captured directly by sensors, yet they determine whether a city is worth living in. A city that is extremely efficient but makes people anxious, isolated, and speechless is not truly smart.

Experiential radiance requires the smart city to include residents' narratives in planning. Questionnaires, interviews, community workshops, participatory maps, observation of children's routes, accompaniment of elderly travel, and night-time safety walks are all methods of manifesting experiential radiance. The city must be able not only to see data, but also to hear life.

Chapter Summary

This chapter separates radiance from landscape aesthetics and returns it to its ontological meaning. Radiance is not lights and screens. It is the entry of urban difference into a state that is visible, perceptible, discussable, and capable of being borne.

Thus, the modes of manifestation in the smart city must be composite: data radiance, spatial radiance, experiential radiance, and public-issue radiance. Digitalization is only one mode of manifestation and cannot monopolize reality.

Implications for Planning and Construction

- Smart city visualization should reduce display-oriented indicators and increase the presentation of real problems and pathways of bearing.
- Urban digital platforms should incorporate residents' experience and field observation, not only machine data.
- Urban public space should be regarded as an interface of radiance rather than the background of a digital system.

Part II

The Urban Vessel: The Bearing Structures of the Smart City

Chapter 5

The Ontology of the Vessel: Manifestation Must Obtain Bearing

Without a Vessel, Radiance Cannot Endure

Radiance is only the beginning, not completion. After a problem is seen, if it lacks a bearing structure, it may more quickly turn into anxiety, anger, and helplessness. Discovering that elderly people have difficulty going up and down stairs without adding elevators, community care, or travel services; discovering waterlogging risk without pipe-network renovation or emergency mechanisms; discovering service shortages without fiscal arrangements or institutional response - all these are cases of seeing without bearing.

The vessel is the structure that allows manifestation to endure and to become order. It may be a road, a building, a platform, an institution, a budget, personnel, law, or a community organization. It may also be algorithmic audit, public participation, and long-term operation mechanisms. The focus of smart city construction must move from the capacity to manifest toward the capacity to bear.

Basic Types of Urban Vessels

Urban vessels include at least six types: spatial vessels, infrastructural vessels, data vessels, institutional vessels, service vessels, and community vessels. Spatial vessels bear daily life. Infrastructural vessels bear movement and energy. Data vessels bear visibility. Institutional vessels bear public decision-making. Service vessels bear concrete needs. Community vessels bear trust and participation.

These vessels cannot replace one another. Technological platforms cannot replace public space. Algorithmic models cannot replace institutional responsibility. Data sharing cannot replace service provision. The urban brain cannot replace community mutual aid. A truly smart city is one in which multiple vessels form stable coordination, not one in which a single vessel monopolizes the city.

The Quality of the Vessel

Not every vessel bears well. A low-quality vessel may appear complete but fail at crucial points. A road that serves only vehicle speed excludes pedestrians. A platform that

serves only departmental assessment excludes residents' experience. An institution that serves only procedural compliance excludes actual response. A community space that serves only display excludes everyday staying.

The quality of a vessel depends on whether it can continuously bear difference, not on whether it looks advanced. When the Luminous City evaluates urban vessels, it asks how much real life they can bear, how many conflicts they can regulate, whether they can retain elasticity under pressure, and whether users are allowed to participate in their correction.

The Danger of the Vessel

The vessel has the danger of becoming self-substantialized. A platform may continually demand data access for the sake of its own completeness. Indicators may compress complex reality for the sake of comparability. Algorithms may exclude heterogeneous experience for the sake of efficiency. Institutions may refuse innovation for the sake of their own safety. At this point, the vessel no longer serves radiance; it overwhelms radiance.

The basic principle of this book is that the vessel must serve radiance, not radiance the vessel. Smart city construction should continually ask: does this system still bear real difference? Does this platform still respond to life? Does this process still protect people? Once the answer is no, the vessel must be repaired, replaced, or reorganized.

Chapter Summary

This chapter proposes the core thesis of the urban vessel: if manifestation cannot enter bearing, it is only the exposure of suffering. The key to the smart city is not whether it can see problems, but whether it can establish sufficiently stable, flexible, and sustainable structures of bearing.

Urban vessels include space, facilities, data, institutions, services, and community. Technology is only one of them. Once any vessel separates itself from difference, it becomes self-substantialized and may in turn oppress radiance.

Implications for Planning and Construction

- Every smart scenario should correspond to a clear bearing vessel: who responds, who invests, who maintains, and who reviews?

- Platform construction must be designed together with spatial transformation, service provision, and institutional procedures.
- For systems with long-term low use and high maintenance, mechanisms of vessel reorganization or exit should be initiated.

Chapter 6

The Spatial Vessel: Roads, Communities, and Public Life

Space Is Not a Neutral Container

Urban space is not a simple container that holds buildings, population, and functions. Space itself organizes difference. It determines who can arrive, who is isolated, who can stay, who is driven away, who can be seen, and who is concealed. Road width, block scale, pedestrian continuity, public-transit coverage, access to green space, and the layout of public facilities all shape urban radiance.

If the smart city ignores space and optimizes only at the data level, it loses the foundation of urban life. An algorithm can optimize vehicle passage but cannot replace a safe sidewalk. A platform can integrate service appointments but cannot replace a community service center within walking distance. The spatial vessel is a basic bearing structure that the smart city cannot bypass.

Roads as Vessels of Movement

Roads bear not only vehicles but also human time, bodies, and rhythms of life. If roads are understood only in terms of traffic efficiency, the city tilts toward motor-vehicle speed, forcing pedestrians, cyclists, children, elderly people, and disabled persons to adapt. True road intelligence should establish a balance among speed, safety, staying, accessibility, and publicness.

The radiance judgment of a road system should not consider only average vehicle speed. It should also consider slow-traffic continuity, crossing safety, bus priority, accessibility quality, the vitality of street interfaces, and thermal comfort. When a road can bear multiple bodies and multiple modes of movement, it becomes a vessel of movement. Otherwise, it is only a privileged channel for one kind of speed.

Communities as Vessels of Life

The community is not the endpoint of an administrative grid, but the front stage of everyday life. Buying groceries, taking children to and from school, elderly people walking, neighborhood care, primary medical services, cultural activities, temporary help, and public deliberation all take place in communities. If the smart city has only city-level

platforms but no community-level bearing, it becomes macroscopically visible but microscopically powerless.

The construction of the community vessel must attend to pedestrian scale, service radius, spatial openness, age-friendliness, child-friendliness, accessibility, neighborly interaction, and the organizational capacity of the community. Smart devices can assist community governance, but they cannot replace community relations. A community platform without community is only a governance terminal. A community space with community is a vessel of life.

Public Space as a Vessel of Commonality

Public space is the place where urban commonality manifests. Squares, parks, street corners, libraries, cultural centers, sports fields, waterfront spaces, and open blocks allow strangers to share the city without having to become the same. The weakening of public space means the weakening of urban public radiance.

Smart city construction should prevent public space from being devoured by commerce, vehicles, landscape projects, and logics of control. Whether a space is smart does not depend on whether it is covered by Wi-Fi or equipped with interactive screens, but on whether it allows people of different ages, incomes, bodily capacities, and cultural backgrounds to stay, encounter, and express themselves with dignity.

Chapter Summary

This chapter explains that space is not the background of technological systems, but an important bearing vessel of urban radiance. Roads, communities, and public spaces determine who can arrive, who can stay, and who can participate in public life.

If the smart city ignores space and optimizes only back-end systems, it creates the urban paradox of an efficient back end and an impoverished front stage. The spatial vessel requires intelligence to return to the bodily scale and everyday life of people.

Implications for Planning and Construction

- Smart transportation evaluation should include the scales of walking, cycling, elderly people, children, and disabled persons.
- Community digitalization should prioritize offline services, public space, and care networks.

- The intelligence of public space should be evaluated by accessibility, the possibility of staying, and the possibility of participation.

Chapter 7

The Data Vessel: From Sensing the City to Interpreting the City

The Meaning of Data Sensing

The data vessel gives the city new sensing capacities. Congestion, energy consumption, pollution, passenger flow, facility failures, and emergency pressure that once could only be judged by experience can now be presented in real time or near real time. Data increases the visibility of complex urban conditions and provides a common language for cross-departmental coordination.

This capacity is especially important for megacities. The larger a city becomes, the harder it is for a single department or individual experience to grasp the whole. The data vessel allows scattered phenomena to enter a common interpretive framework, enabling the city to move from knowing after the event to early warning, and from local experience to systemic observation.

Data Is Not Interpretation Itself

Data can describe phenomena, but it does not automatically provide meaning. Increased foot traffic in an area may mean higher vitality, or it may mean transfer congestion caused by insufficient public transportation. Fewer complaints may mean that problems have been solved, or it may mean that residents no longer believe feedback is useful. More police reports may mean higher risk, or it may mean easier reporting channels.

Therefore, the smart city must move from sensing the city to interpreting the city. Data must be combined with field investigation, institutional background, residents' experience, and historical context. Data without interpretation is only a fragment of radiance. Data that has been interpreted can enter decisions of bearing.

Data Silos and the Fracture of Radiance

Data silos are not only a technical problem but also an ontological problem. When data on transportation, urban management, public security, emergency response, housing and construction, environmental protection, civil affairs, and communities are separated from one another, urban radiance is cut apart by departmental boundaries. Each system sees a part, but no one sees the relations.

The Luminous City needs data coordination, but coordination is not borderless sharing. Data sharing must follow legitimate purpose, minimum necessity, clear authority, auditable security, and privacy protection. Otherwise, breaking data silos may become data overflow and in turn damage residents' trust.

The Ethical Task of Data Governance

Data governance usually focuses on standards, quality, security, sharing, and interfaces. This book further emphasizes that data governance also bears an ethical task of manifestation: it must ensure that urban differences are appropriately expressed rather than misclassified, oversimplified, or arbitrarily concealed.

For example, vulnerable groups should not be coded only as service objects; they should also be recognized as participants in public decision-making. Environmental risks should not be treated only as early-warning points; they should also be understood as ecological bearing boundaries. Residents' complaints should not be counted only as case-closure rates; they should also be read as signals of urban will continuing to manifest.

Chapter Summary

This chapter emphasizes the duality of the data vessel. Data can expand urban sensing, but it can also produce new concealments. Data is not interpretation itself; it must be combined with field conditions, history, institutions, and residents' experience.

The goal of data governance is not only sharing and security, but ensuring that urban difference is appropriately expressed. Data silos fracture radiance, while data overflow may damage trust.

Implications for Planning and Construction

- When building data platforms, cities should design data-interpretation mechanisms at the same time, rather than pursuing only the quantity of data connections.
- Important indicators should be accompanied by interpretive notes that explain possible misreadings and boundaries of application.
- Data sharing should follow principles of purpose limitation, minimum necessity, and auditability.

Chapter 8

The Algorithmic Vessel: Intelligence or Concealment

The Manifestational Capacity of Algorithms

Algorithms can discover patterns in complex data, predict risks, optimize dispatching, and assist resource allocation. Traffic-signal optimization, disaster-risk warning, public-service demand prediction, energy-load management, facility-failure identification, and emergency-response ordering can all use algorithms to improve efficiency. The algorithmic vessel therefore becomes an important component of the smart city.

In the Luminous City, the value of algorithms lies in enhancing the city's capacity to sense and bear difference, not in replacing public judgment. Algorithms should help the city see risks earlier, allocate resources more reasonably, discover imbalances faster, and support human decisions more effectively.

The Danger of Algorithmic Concealment

Algorithms can also create new concealments. Training data may contain historical biases. Classification models may exclude nonstandard lives. Prediction systems may misread poverty, disease, or mobility as risk. Black-box decisions may leave affected persons without appeal. The more trusted an algorithm becomes, the harder its concealments are to discover.

If the smart city treats algorithmic results as neutral facts, it allows bias to re-enter governance in technical form. Algorithms cannot be exempt from explanation because they are complex, exempt from scrutiny because they are efficient, or allowed to ignore severe harm to minorities because they are statistically accurate.

Algorithms Are Not Urban Will

Algorithms can calculate schemes, but they cannot determine value for the city. Whether to prioritize vehicle speed or pedestrian safety, whether to increase security density or protect privacy, and whether to prioritize commercial vitality or community quiet are not purely technical questions. They are judgments of public value.

Therefore, algorithms are not urban will. They are vessels. Urban governance cannot hand responsibility over to models, nor can it replace political, ethical, and planning

judgment with system recommendations. The Luminous City requires algorithms always to remain within a public framework that is explainable, contestable, and correctable.

Algorithmic Audit and Radiant Ethics

Algorithmic audit is an important mechanism by which the smart city avoids alienation. Auditing must examine not only model performance, but also data sources, variable selection, boundaries of application, consequences of misjudgment, affected groups, paths of correction, and attribution of responsibility.

Radiant ethics requires every algorithm to answer four questions: what does it manifest? What does it conceal? Whom does it exclude? Who bears the harm it causes? Only algorithms that can answer these questions deserve to become urban vessels.

Chapter Summary

This chapter positions algorithms as urban vessels, not urban subjects. Algorithms can enhance prediction, dispatching, and identification, but they cannot decide value for the city or assume responsibility in place of governors.

The greatest danger of algorithms is that they reproduce bias in the form of technological neutrality and weaken appeal and explanation through black-box mechanisms. The Luminous City must place algorithms within a public framework that is explainable, auditable, and correctable.

Implications for Planning and Construction

- All algorithms involving public resources, rights, opportunities, and risk labels should be subject to algorithmic auditing.
- Algorithm procurement should require explainability, descriptions of application boundaries, and correction mechanisms.
- Urban decisions must not cancel public discussion and responsibility by invoking system recommendations.

Chapter 9

The Institutional Vessel: Planning, Governance, and Public Decision-Making

Institutions Carry Manifestation into Response

Technology can discover problems, but it cannot automatically solve them. If a platform discovers insufficient public services in an area, planning adjustment, fiscal arrangement, departmental coordination, project construction, and long-term operation are still required. The role of the institutional vessel is to translate manifested difference into public response.

Without institutional bearing, the smart city remains at the stage of seeing the city. No matter how advanced an urban operation center is, if discovered problems cannot enter budgets and projects, if predicted risks have no responsible actor, and if abundant residents' feedback lacks mechanisms of negotiation and handling, true intelligence cannot be formed.

Planning Institutions as Pathways of Manifestation

Territorial spatial planning, special planning, detailed planning, urban renewal, resilient city construction, public-service facility allocation, and urban health checks are all pathways through which urban differences enter formal institutions. GB/T 36333-2018 emphasizes the overall principles, basic processes, demand analysis, overall design, architecture design, and implementation paths of top-level smart city design. This shows that the smart city itself also requires institutionalized design.

The Luminous City should understand top-level design as the design of manifestational architecture, not merely as the design of technical architecture. Top-level design must answer: how are problems discovered? How are differences classified? How is data interpreted? How does the public participate? How are projects transformed? How does feedback return?

Governance Processes as Urban Vessels

Hotlines, grids, government services, comprehensive law enforcement, emergency linkage, grassroots councils, community workstations, and digital platforms together form vessels of governance process. They determine whether urban radiance can move from

discovery to handling, from handling to improvement, and from improvement to institutional learning.

Good governance processes are not only about traceability, nor are they simply about increasing case-closure rates. They must identify the structural causes behind recurring problems, transform individual complaints into planning clues, translate grassroots experience into institutional adjustment, and turn residents' dissatisfaction into a motive force for urban reorganization.

The Risk of Institutional Rigidity

Institutions can also become exhausted. When institutions pursue only procedural correctness, avoidance of responsibility, and completeness of forms, real problems are devoured by procedure. Grassroots personnel become fatigued by platform reporting, residents become disappointed by repeated submission of materials, and departments become unwilling to experiment because they fear risk. The urban vessel begins to stiffen.

The Luminous City requires institutions to remain revisable. An institution is not a perfect vessel formed once and for all, but a bearing structure capable of continuous learning. Every institution that cannot respond to new differences should be re-examined. Every institution that turns people into appendages of process must be repaired.

Chapter Summary

This chapter points out that after technology discovers problems, discovery must enter institutional response. Without planning, finance, departmental coordination, procedural responsibility, and public participation, data manifestation cannot become urban improvement.

The institutional vessel can also become rigid. It may turn from a structure of bearing into a process machine. The Luminous City requires institutions to retain learning capacity and to transform individual problems into structural repair.

Implications for Planning and Construction

- Top-level smart city design should also design the paths by which problems enter planning and budgeting.
- Hotline, grid, and platform data should be regularly transformed into lists for urban renewal and public-service optimization.

- Institutional assessment should reduce purely trace-based indicators and increase indicators of real improvement.

Chapter 10

The Community Vessel: Residents, Communities, and Urban Publicness

The City Has More Than a Management Side

Smart city projects often begin from the government side, the management side, and the platform side, emphasizing urban operation, departmental coordination, and event handling. These are necessary. But if the city has only a management side, residents are fixed as objects: served, counted, warned about, and governed.

The Luminous City must also have a resident side, a community side, and a public-deliberation side. Residents are not only data sources, nor are they objects of governance. They are participants in urban manifestation and reorganization. Urban intelligence should not exist only in the back end; it should also exist in public discussion, community mutual aid, and residents' action.

The Irreplaceability of Residents' Experience

Residents know which road feels frightening at night, which intersection is dangerous for children crossing, which compound makes it hard for elderly people to go downstairs, which green space is never truly used, and which service appears to exist but is hard to obtain. Such experience is often closer to the scene of the problem than macro-level data.

If the smart city does not absorb residents' experience, it produces a state of governance that is clear at a distance and blurred nearby. Every point on the platform is clear, while inconveniences in life remain unintelligible. The value of the community vessel is that it allows experiential radiance to enter urban interpretation.

Public Participation Is Not Decoration

Public participation should not be merely a formal stage of hearings, questionnaires, and notices. It should become an institutional mechanism through which urban difference enters manifestation, judgment, and bearing. The key to participation is not to have residents endorse a predetermined scheme, but to allow the scheme to be corrected by residents' experience.

Effective participation requires transparency of information, language that is understandable, channels that are accessible, feedback that is visible, and results that can be tracked. Otherwise, participation becomes ritual and further consumes public trust. The Luminous City treats participation as a community vessel, not as publicity decoration.

Community Radiance

Community radiance does not mean that everyone agrees. It means that different people can see, recognize, and negotiate with one another in the city. It appears as trust among neighbors, a sense of safety in public space, community care for the vulnerable, residents' participation in urban affairs, and the capacity to handle public conflicts rationally.

A smart city without community radiance may be efficient but cold, precise but alienated, safe but unfree. A true Luminous City is one in which technology, space, institutions, and community shine together.

Chapter Summary

This chapter restores residents and communities from objects of governance to participants in urban radiance. Without residents' experience, the smart city can possess distant clarity while becoming blind nearby.

The community vessel allows the city not only to be managed but also to be lived together. It bears trust, mutual aid, negotiation, and publicness, and it is a form of urban wisdom that technological systems cannot replace.

Implications for Planning and Construction

- Smart cities should establish resident-side mechanisms for feedback, deliberation, and result tracking.
- Community digitalization cannot replace community organization and offline public space.
- Public participation should enter before schemes are finalized, not merely endorse them after completion.

Part III

Topology and Admissibility: A Boundary Theory of the Smart City

Chapter 11

Urban Topology: Nodes, Flows, Boundaries, and Levels

The City Is Not a Flat Map

The city is not a uniformly spread plane, but a topological structure composed of nodes, flows, boundaries, levels, and gaps. The same distance means different accessibility under different transport conditions. The same facility means different opportunities of use across different social boundaries. The same data platform means different response capacities at different institutional levels.

A topological perspective allows the smart city to look not only at points and quantities, but at relations. How many cameras, sensors, or service points a city has does not equal a better city. The key is how these points are connected, whether connection is accessible, whether accessibility is fair, and whether fairness can continue.

Nodes

Transport hubs, commercial centers, hospitals, schools, community centers, government-service centers, data centers, cultural facilities, and emergency nodes are all places where urban radiance gathers. The stronger a node is, the more it attracts flows. But excessive concentration of nodes can also create congestion, exclusion, and regional imbalance.

The smart city must understand the duality of nodes. A node is both a vessel of bearing and a source of pressure. Planning cannot pursue only node strength. It must also attend to coordination among nodes, the effects of nodes on surrounding communities, and whether weak nodes lead some areas into long-term low manifestation.

Flows

The city is composed of many kinds of flows: people, vehicles, logistics, information, capital, energy, services, ecology, and risk. Smart city data systems are often best at capturing flows, but planning needs to understand the conflicts among flows.

For example, commercial flows may pressure residential quiet. Motor-vehicle flows may cut off children's routes to school. Logistics efficiency may increase neighborhood noise. Concentrated information flows may produce platform monopolies. Risk flows may

spread along drainage, transportation, and social vulnerabilities. The Luminous City must regulate flows rather than blindly accelerate them.

Boundaries and Levels

Urban boundaries are not only administrative boundaries. There are also service boundaries, psychological boundaries, digital boundaries, ecological boundaries, class boundaries, and cultural boundaries. Many urban problems occur at boundaries: between urban and rural areas, old city and new district, commercial area and residential area, online service and offline service.

Levels are equally important. Household, building, community, district, urban area, metropolitan region, and regional network are nested within one another. Optimization at one level may produce damage at another. If the smart city lacks level-awareness, it may mistake local efficiency for overall wisdom.

Chapter Summary

This chapter uses a topological perspective to show that the city is not a collection of points on a plane, but a relational structure composed of nodes, flows, boundaries, and levels. The smart city must understand relations, not merely count quantities.

Topological analysis can reveal the relations between center and periphery, flow and obstruction, service and exclusion. It allows planners to see how urban radiance is distributed and where it fractures.

Implications for Planning and Construction

- Smart city maps should express accessibility, boundaries, and level relations, not only facility points.
- Flow optimization should evaluate impacts on surrounding communities, ecology, and vulnerable groups.
- Problems across administrative and departmental boundaries should have specialized coordination mechanisms.

Chapter 12

Admissibility: The Thresholds of Urban Bearing

The Meaning of Admissibility

Admissibility refers to the boundary conditions under which urban vessels can bear difference without collapsing, becoming alienated, or becoming exhausted. It is not a conservative restriction, but the precondition for the city to continue manifesting. Without boundaries, every bearing structure will be overused, and every efficiency will turn into consumption.

In the Luminous City, admissibility is a core category of planning. It reminds us that roads have capacity, ecology has thresholds, finance has boundaries, communities have degrees of endurance, privacy has inviolable space, grassroots governance has limits of fatigue, and human attention and trust are not infinite resources.

Transport and Ecological Admissibility

Transport admissibility shows that more urban movement is not always better. Excessively long commutes, overdependence on motor vehicles, inconvenient transfers, fractured slow-traffic systems, and normalized congestion indicate that movement has exceeded the boundaries of its bearing vessels. Smart transportation should not only increase passage speed; it should also reduce unnecessary movement, optimize the radius of daily life, and improve public transit and slow-traffic systems.

Ecological admissibility is equally crucial. Heat islands, waterlogging, air pollution, water-resource pressure, insufficient green space, and fragmented biological habitats are signals that ecological vessels have been breached. If the smart city pursues operational efficiency while ignoring ecological boundaries, it hides future risks inside present efficiency.

Social and Governance Admissibility

Social admissibility involves housing costs, commuting time, service accessibility, care pressure, community stability, and public trust. When the cost of living continues to rise, public services remain insufficient, and residents are forced to sacrifice family and health for urban opportunity, the city is at the edge of social exhaustion.

Governance also has admissibility. Too many platforms, forms, assessments, and overly long chains of responsibility can make grassroots governance lose real response capacity. Smart governance cannot be achieved at the cost of endlessly squeezing the grassroots. Otherwise, the governance system appears refined on the surface while being exhausted internally.

Data and Privacy Admissibility

Data collection also requires boundaries. Public safety, traffic management, and service optimization may require data, but this does not mean that the city may collect, retain, associate, and predict without limit. Privacy is the right of a person to maintain an unmanifest space and a boundary of personality.

The Luminous City recognizes the importance of manifestation, but it also recognizes the right not to be manifested. The smart city must establish ethical boundaries between visibility and concealment: public problems need manifestation, private life needs protection; risks need warning, ordinary life should not be continually suspected.

Chapter Summary

This chapter proposes admissibility as the boundary theory of the smart city. Urban vessels do not bear without limit. Transport, ecology, finance, governance, social trust, and privacy all have thresholds.

True intelligence is not unlimited optimization, but knowing where to stop, buffer, divert, protect, and yield. Without admissibility, efficiency becomes exhaustion.

Implications for Planning and Construction

- Smart city construction should establish threshold indicators for transport, ecology, finance, governance load, and privacy.
- When thresholds are exceeded, priority should be given to decompression and reorganization rather than to adding more technology.
- Data collection should be included within admissible boundaries in order to prevent unlimited association and permanent retention.

Chapter 13

Smart City Resilience: Maintaining Manifestation under Shock

Resilience Is Not Restoration to the Original State

The resilient city is often understood as the capacity to recover after disaster. In the ontology of radiance, however, resilience is not simply a return to the original state. It is the capacity to maintain key capacities of manifestation, bearing, and reorganization under shock. If the original state is already fragile, restoring it is not resilience but the repetition of risk.

True resilience means that when the city encounters floods, epidemics, fires, power outages, extreme heat, or social conflict, it can still see problems, protect life, mobilize resources, maintain basic services, and complete structural learning afterward.

Infrastructure Resilience

Electricity, communication, water supply, drainage, gas, transportation, medical care, and emergency systems are the underlying bearings of continuous urban radiance. In ordinary times they are often invisible; only during failures and disasters do they reveal their importance. The smart city should improve infrastructure resilience through monitoring, redundancy, distributed configuration, and cross-system linkage.

But resilience cannot rely only on large systems. The more complex large systems become, the more likely they are to suffer cascading failures under extreme conditions. The Luminous City needs both strong systems and weak ties: efficient city-level dispatching as well as community-level self-organization, local knowledge, and temporary mutual aid.

Community Resilience

When disaster occurs, the earliest responses often come from neighbors and communities. Which household has elderly people, which road floods easily, where one can take temporary shelter, who needs medicine, who has tools or vehicles - such information is often not on the urban screen, yet it is crucial for protecting life.

Community resilience is therefore the foundation of smart city resilience. Community organizations, volunteer networks, public spaces, primary medical care, emergency supplies, and residents' trust together constitute a community vessel under shock. A smart city without community resilience is a city with a neural center but without capillaries.

Resilience and Reorganization

Resilience must ultimately enter reorganization. Every disaster and crisis is a moment when urban vessels are tested. Where systems failed, where congestion occurred, where response was slow, and where vulnerable people were missed should all become bases for institutional and spatial renewal.

The Luminous City does not treat disaster merely as an exceptional event. It treats disaster as the forced manifestation of hidden differences. Disaster exposes unmanifest vulnerabilities; the responsibility of planning is to transform this exposure into future bearing capacity.

Chapter Summary

This chapter understands resilience as the capacity for manifestation, bearing, and reorganization under shock, not as simple restoration to the original state. If the original state is fragile, restoring it only repeats risk.

Resilience requires both large infrastructure and community organization, local knowledge, and neighborly mutual aid. The resilience of the Luminous City is the combination of systemic capacity and community capacity.

Implications for Planning and Construction

- Smart cities should establish post-disaster review mechanisms that turn shocks into updates of urban vessels.
- Resilience planning should cover both infrastructure redundancy and community self-organization.
- Risk warning should pay special attention to vulnerable groups such as elderly people, children, and disabled persons.

Part IV

Exhaustion, Alienation, and Concealment: The Reverse Side of the Smart City

Chapter 14

The Exhausted City: When Smart Systems Consume the City They Were Meant to Serve

Technological Exhaustion

Technological exhaustion is a common reverse form of the smart city. More and more systems are built, more interfaces are connected, more applications are deployed, and maintenance costs continue to rise, while actual use and governance improvement remain limited. Platforms become platforms for their own sake; projects become projects for their own sake. Technology no longer bears difference but begins to consume finance, personnel, and attention.

The mark of technological exhaustion is not that there is too little technology, but that technology is excessive while bearing remains insufficient. Grassroots workers repeatedly enter data into multiple systems; residents repeatedly authenticate themselves on multiple platforms; departments continually report data for display. At this point, the smart city has not reduced urban burdens but has created new ones.

Fiscal Exhaustion

Smart city construction often emphasizes construction and neglects operation. One-time construction funds can support platforms, devices, and exhibition halls, but long-term maintenance, data governance, personnel training, system iteration, and security audits require continuous investment. Without sustainable fiscal arrangements, urban vessels quickly age.

Fiscal exhaustion is not simply a lack of money, but an imbalance between investment and bearing. If a high-cost system cannot solve real problems, it squeezes out more direct vessels of bearing such as education, medical care, community services, and infrastructure renewal. The Luminous City requires every smart investment to undergo the test of bearing capacity.

Governance Exhaustion

Governance exhaustion appears when grassroots levels are overwhelmed by platforms, assessments, reports, check-ins, and trace requirements. Smart governance originally seeks to improve coordination, but it can turn the grassroots into a machine for

data production. The event has not yet been resolved, but the process has been completed. Residents are not yet satisfied, but the system has closed the case.

This exhaustion is especially dangerous because it creates the illusion of governance. The screen shows procedural closure while the real problem circulates again. Indicators show rapid response while residents experience sluggishness. The Luminous City must regard grassroots governors as people who also need to be borne, not as institutional parts that can be exploited without limit.

Exhaustion of Meaning

The city may also suffer exhaustion of meaning. It becomes more efficient but less warm; more visible but harder to dwell in; safer but less free; more convenient but more homogeneous. People complete tasks in the city but no longer gain belonging and memory.

The exhaustion of meaning shows that urban radiance has been covered over by functionality. The smart city should not ask only how to become faster, more accurate, or more economical. It should also ask whether life is richer, whether relationships are more reliable, whether public spaces are more open, and whether residents can still treat the city as their own lifeworld.

Chapter Summary

This chapter discusses the reverse side of the smart city: technology, finance, governance, and meaning can all be consumed by the smart system itself. More systems and more complex platforms do not necessarily mean stronger bearing.

Exhaustion reminds us that construction is not the end. Operation, maintenance, learning, and exit are equally important. A smart system that cannot continuously bear becomes a new burden on the city.

Implications for Planning and Construction

- Smart project initiation should include assessment of full life-cycle costs and exit mechanisms.
- The burden placed on grassroots users should be an important evaluation indicator for smart governance systems.

- High-cost projects with low improvement should be adjusted, merged, or stopped in time.

Chapter 15

The Concealed City: Who Is Not Seen by the Smart City

People Outside Data

The smart city most easily sees high-frequency, standardized, locatable, and quantifiable people and behaviors. It less easily sees low-frequency, nonstandard, marginal, and silent people. Elderly people, children, disabled persons, migrants, people without smart devices, and those with weak digital capacity may be precisely those who need the city's support most.

Concealment is not always malicious. It often comes from default assumptions in system design: assuming that everyone has a phone, can use an app, can read notices, can make online appointments, and is willing to leave data. These defaults silently exclude some people from smartness.

Experience Outside Platforms

Many urban experiences cannot easily enter platforms: fatigue, loneliness, fear, humiliation, helplessness, disappointment, the rupture of intimate relations, and the disappearance of local memory. They do not necessarily form complaints, nor do they necessarily correspond to one responsible department, but they deeply affect the quality of urban life.

The Luminous City must seek expressive forms for these experiences. Social surveys, urban narratives, participatory observation, community archives, oral history, and public art can all become modes by which non-platform experience is manifested. Intelligence should not belong only to what machines can read.

The City Outside Indicators

A city may perform well on indicators and still feel alienating. Service coverage may be high while services are hard to obtain. Green-space area may meet standards while green space is not accessible on foot. Platform response may be rapid while recurring problems do not decrease. Safety indexes may improve while public spaces lose openness.

This shows that the city continues to exist beyond indicators. Evaluation of the Luminous City must preserve a sense of the scene, a bodily sense, and a public sense.

Indicators should become entrances to discovering problems, not exits for concealing them.

The Consequences of Concealment

Concealed differences do not disappear. They remanifest as conflict, disease, migration, decline, distrust, spatial decay, or disaster loss. The more unwilling the city is to see them, the more intense their return may become.

Concealment is therefore not a static defect but a future risk. If the smart city cannot actively manifest low voices, it will be forced to face high-intensity eruptions. The governance wisdom of the Luminous City lies in hearing weak signals in advance.

Chapter Summary

This chapter shows that the smart city may see more, but it may also conceal some people more precisely. Low-data existence, experience outside platforms, and life outside indicators constitute the parts of the city most easily lost.

Concealment does not make difference disappear; it only allows difference to return in more violent ways. The Luminous City should actively hear weak signals rather than wait for risks to erupt.

Implications for Planning and Construction

- Smart systems should undergo accessibility tests for vulnerable groups, including offline alternative channels.
- Urban evaluation should include narrative materials, field visits, and indicators of residents' feelings.
- Areas with low complaint numbers should not be directly judged as low-problem areas; trust and expressive capacity should also be analyzed.

Chapter 16

The Alienated City: When the Vessel Overwhelms Radiance

Platform Alienation

Platforms should serve the manifestation and bearing of urban difference. But when a platform takes its own completeness, display, and controllability as its goal, it begins to alienate. Access for the sake of access, display for the sake of display, and closure for the sake of closure: the platform changes from a vessel into the center.

The result of platform alienation is that reality obeys the system rather than the system responding to reality. Grassroots workers must break complex problems into platform options; residents must translate life difficulties into standard work orders; departments must convert actual work into displayable data. Radiance is compressed into formats the platform can recognize.

Algorithmic Alienation

Algorithmic alienation occurs when model results are treated as unquestionable facts. People no longer discuss why a judgment is made, but only what result the system gives. They no longer ask about value trade-offs, but only accuracy rates. They no longer assume responsibility, but say that the algorithm recommended this.

Such alienation weakens public judgment. Urban governance needs reasons, responsibility, and explanation, not only probability. The Luminous City allows algorithms to provide assistance, but it does not allow algorithms to monopolize judgment. Every algorithm that affects rights, resources, and life opportunities must accept public scrutiny.

Spatial Alienation

Spatial alienation appears when urban space serves capital display, vehicle speed, tourist consumption, or landscape performance rather than everyday life. Streets become channels, squares become exhibition grounds, historic districts become consumption fields, community space becomes management grids. People are allowed to pass through but not encouraged to stay.

When space cannot bear ordinary life, urban radiance becomes superficial. Smart devices can make space look modern, but they cannot replace real spaces where people can sit, walk, play, communicate, and remember.

Governance Alienation

Governance alienation appears as an excess of formal rationality and an insufficiency of life-bearing. Every matter has a process, every link has a trace, every responsibility is decomposed, yet the real problem remains hard to solve. Institutions maintain their own safety while urban life is forced to adapt to them.

The Luminous City must oppose this alienation. The goal of governance is not to prove that the system is correct, but to make the city more capable of bearing life. If an institution can produce compliance but not improvement, it needs to be reorganized.

Chapter Summary

This chapter points out that when platforms, algorithms, spaces, and institutions no longer serve urban difference but instead serve their own expansion, display, and safety, the vessel becomes alienated.

The essence of alienation is that the bearing structure requires life to adapt to itself. The Luminous City must continually bring the vessel back to human life, public value, and sustainable bearing.

Implications for Planning and Construction

- Platform design should allow nonstandard problems to enter instead of forcing life to adapt to fixed forms.
- Spatial renewal should evaluate whether original residents, everyday life, and local memory are being displaced.
- Institutional processes should aim at problem improvement, not at process self-proof.

Part V

Radiant Ethics: Value Constraints for the Smart City

Chapter 17

Manifestation Is Not Identical with the Good

Strong Manifestation Can Also Be Oppression

Manifestation does not automatically equal the good. A city that is highly monitored, highly predictive, highly visible, and highly linked may be a city of strong manifestation, but it is not necessarily a good city. It may turn every person into a traceable object, every deviation into a risk, and every heterogeneous form of life into an abnormality.

Radiant ethics must therefore constrain manifestation. It is not better simply to see more. One must ask why something is seen, who sees it, how what is seen is used, whether those who are seen have rights, and whether there are private spaces that should not be manifested.

The Good Does Not Lie in the Intensity of Manifestation

The good of the city does not depend on the intensity of manifestation, but on the mode of bearing after manifestation. Is poverty seen in order to provide assistance or in order to expel? Is gathering seen in order to serve or to suppress? Is risk seen in order to protect or to label? Are the vulnerable seen in order to respect them or to manage them?

The ethical judgment of the Luminous City must move from what the city sees to how the city treats those who are seen. Manifestation without bearing and respect may be only more precise domination.

Basic Questions of Radiant Ethics

Every smart city project should answer seven questions: whom does it manifest? Whom does it conceal? What does it bear? What does it consume? What does it protect? Whom does it exclude? Does it make human life more dignified? These questions form the basic checklist of radiant ethics.

This checklist is not a moral appendix. It should enter project initiation, scheme review, data governance, algorithm procurement, planning evaluation, and operational review. Without ethical review, the smart city may produce new injustice in the name of efficiency.

Against Technological Fear and Technological Worship

Radiant ethics opposes both technological worship and technological fear. Technology is neither a savior nor a natural enemy. Its value depends on the kind of urban vessel into which it is organized, the kind of difference manifestation it serves, and the boundary constraints it accepts.

The position of this book is therefore critical use of technology. Technology should become an extension of people, not a replacement for them; a tool of the urban community, not its ruler; a vessel of radiance, not a devourer of radiance.

Chapter Summary

This chapter establishes the bottom line of radiant ethics: manifestation is not identical with the good, and strong manifestation can also be oppression. Whether a city is good does not depend on how much it sees, but on how it treats those who are seen.

Radiant ethics is not anti-technology. It requires technology to accept value constraints. Manifestation must coexist with dignity, bearing, rights, boundaries, and responsibility.

Implications for Planning and Construction

- Smart city projects should establish an ethical review checklist.
- Projects involving personal data and automated decision-making should provide appeal and correction paths.
- All tools of manifestation should state whom they protect and whom they might harm.

Chapter 18

Human-Centeredness: How Human Difference Becomes the Scale of the City

People Are Not Abstract Users

The smart city often calls people users, population, traffic, service objects, event subjects, or risk factors. These terms make management easier, but they can erase human bodies, emotions, memories, vulnerabilities, and dignity. Human-centeredness first means opposing the reduction of people to system variables.

Human beings exist with difference. Children, elderly people, pregnant women, disabled persons, night-shift workers, migrant workers, caregivers, low-income families, new citizens, and tourists do not have the same urban needs. The scale of the city must move from the abstract average person to concrete differentiated persons.

The Scale of Children

A child-friendly city is not merely a city with more children's facilities. It is a city in which children can explore safely. Safe routes to school, accessible play spaces, street corners where one can stay, slow-traffic-friendly communities, understandable signage, and permitted play are all forms of urban radiance at the scale of children.

Starting from children, the city exposes many problems ignored by adult perspectives: excessive vehicle speed, difficult crossings, over-commercialized public space, lack of contact with nature in communities, and lack of relations of care on streets. The scale of children allows the city to relearn softness and safety.

The Scale of Aging and Accessibility

An age-friendly city does not label elderly people as special. It recognizes aging as a difference of life that everyone will experience. Elevators, ramps, seats, shade, community medical care, slow-traffic safety, offline services, and understandable information are all age-friendly vessels.

Accessibility is not an additional need of a minority, but a basic civilization that follows from the city's recognition of bodily difference. Ramps, tactile paving, accessible toilets,

low service counters, auditory and visual assistance, and continuous passage routes ensure that the city is not designed only for the standard body.

The Scale of Ordinary Life

Ultimately, the smart city must return to ordinary life: whether one can buy groceries conveniently, walk safely, seek medical care smoothly, commute reliably, find a seat, avoid storms, protect privacy, participate in the community, obtain help, and preserve memory.

If high technology cannot improve these forms of ordinary life, it is suspended intelligence. The Luminous City does not bring people into the system; it brings the system back to people.

Chapter Summary

This chapter concretizes human-centeredness as the scale of differentiated persons. A human being is not an abstract user, but an existence with age, body, memory, emotion, vulnerability, and dignity.

The scales of children, elderly people, disabled persons, and ordinary life bring the smart city back from abstract indicators to bodily experience. Urban intelligence must withstand the test of the most vulnerable body.

Implications for Planning and Construction

- Smart city design should conduct scenario tests for children, elderly people, and disabled persons.
- Online services must retain offline accessible paths in order to avoid digital exclusion.
- Planning evaluation should include a test of convenience during an ordinary day.

Chapter 19

The Boundaries of Freedom, Privacy, and Public Safety

Safety Cannot Justify Unlimited Manifestation

Public safety is a basic responsibility of the city. Disaster warning, public security prevention, fire management, public health, and emergency response all require some degree of sensing and coordination. But safety cannot justify unlimited manifestation. If all life is continuously monitored because of potential risk, the city shifts from protecting life to oppressing life.

The Luminous City must distinguish necessary manifestation from excessive manifestation. Public risks need to be manifested, while private life should not be exposed without distinction. Dangerous behavior needs warning, while ordinary difference should not be suspected. Emergency states may require temporary intensification, but normal life cannot permanently enter emergency logic.

Privacy as the Right to the Unmanifest

Privacy is not a backward obstacle. It is a person's right to the unmanifest. Everyone needs space in which not to be observed, recorded, predicted, or evaluated. Without such space, people lose the boundary of personality within the city.

The smart city should treat privacy as part of urban admissibility. Data minimization, purpose limitation, anonymization, access control, retention limits, user knowledge, exit mechanisms, and independent audit are institutional vessels that protect the right to the unmanifest.

Freedom as the Condition for the Generation of Difference

Urban freedom is not disorder. It is the condition that allows new ways of life, new public relations, new cultural expressions, and new urban possibilities to emerge. A completely predictable city may also be a city that has stopped generating.

If the smart city excessively pursues behavioral standardization and risk elimination, it compresses urban creativity. Street performances, temporary markets, night activities, community self-governance, youth culture, and public discussion all require a certain uncertainty. Freedom is the condition by which difference continues to generate.

Boundary Balance

There is no once-and-for-all answer among safety, efficiency, freedom, privacy, dignity, and publicness. They must be carefully balanced in different situations. The task of the Luminous City is not to swallow all other values with one value, but to establish boundary structures that are explainable, reviewable, and correctable.

Smart cities should therefore establish clear norms for data ethics, mechanisms for algorithmic governance, procedures for public participation, and paths of rights remedy. The clearer the boundaries, the more trustworthy the intelligence.

Chapter Summary

This chapter explains that public safety cannot become a reason for unlimited manifestation. Privacy is the right to the unmanifest, and freedom is the condition for the continued generation of difference.

The value difficulty of the smart city lies in balancing safety, efficiency, freedom, privacy, and dignity. The clearer the boundaries, the more trustworthy the intelligence.

Implications for Planning and Construction

- Urban data collection should follow purpose limitation, minimum necessity, and time limits.
- Public safety systems should distinguish ordinary governance from emergency states and avoid normalizing emergency logic.
- Residents should have basic paths for knowledge, query, correction, and appeal.

Part VI

Reorganization and Renewal: The Methodology of Building the Luminous City

Chapter 20

Urban Health Checks: Finding Unmanifest Wounds

The Ontological Meaning of Urban Health Checks

Urban health checks are not merely administrative evaluations, but processes for finding unmanifest urban wounds. In recent years, the field of housing and urban-rural development has emphasized the use of urban health checks as an important lever for coordinating planning, construction, and management, and for linking with urban renewal. This is internally consistent with the logic of the Luminous City.

Health checks matter because a city does not automatically present its own problems to planners. Many risks hide within everyday endurance, many inconveniences are treated as individual difficulties, and many structural wounds become visible only through systematic observation.

Multi-Source Sensing

A health check for the Luminous City should combine multi-source sensing: remote sensing, the Internet of Things, traffic data, environmental data, facility records, public-service data, hotline complaints, community interviews, field surveys, residents' maps, and grassroots records. A single source will be biased; multi-source corroboration comes closer to reality.

It is especially necessary to combine data sensing with bodily experience. Sensors can discover waterlogging points, but they may not know how far residents detour. Platforms can show service coverage, but they may not know whether elderly people can make appointments. Maps can mark green spaces, but they may not know whether those spaces are safe, comfortable, and accessible.

From Problem Lists to Bearing Lists

Urban health checks cannot stop at problem lists. After problems are discovered, they should further form bearing lists: which problems are borne by spatial renewal, which by institutional adjustment, which by service expansion, which by data governance, which require continuous fiscal input, and which require public participation.

This step is crucial to the Luminous City. If the health check only produces a report, the report becomes a new concealment. Health checks must enter planning, projects,

budgets, operations, and review in order to transform unmanifest wounds into urban reorganization.

The Periodicity of Health Checks

Urban health checks should be periodic, because urban differences change, old problems shift, new problems arise, some vessels age, and some measures produce side effects. A single health check can only describe the radiant state of one time slice; it cannot replace continuous observation.

The Luminous City treats the health check as a mechanism of urban self-sensing. It is not for proving that the city is perfect, but for allowing the city continuously to discover its blind spots.

Chapter Summary

This chapter understands the urban health check as a mechanism for finding unmanifest wounds. The health check is not to prove urban achievements, but to discover blind spots and transform them into bases for reorganization.

Urban health checks must use multi-source corroboration and move from problem lists to bearing lists. Otherwise, the health-check report itself may become a new formalism.

Implications for Planning and Construction

- Urban health checks should include data indicators, field surveys, and residents' experience at the same time.
- Results of health checks should enter project libraries, budget libraries, and urban renewal plans.
- Health checks should be periodic, and mechanisms should be established for tracking problem improvement.

Chapter 21

Top-Level Design: From Technical Architecture to Manifestational Architecture

Rethinking Top-Level Design

Top-level smart city design usually includes goals, architecture, data, platforms, applications, standards, security, and implementation paths. GB/T 36333-2018 has already provided basic processes and recommendations for top-level design. On this basis, this book proposes that the deeper task of top-level design is to design the architecture of urban difference's manifestation, bearing, and reorganization.

In other words, top-level design should not begin by drawing a technical system diagram. It should first clarify which urban problems most need to be seen, which differences are persistent, which bearing vessels are insufficient, which boundaries are being breached, and which forms of exhaustion have already appeared.

Manifestational Architecture

Manifestational architecture includes mechanisms of problem discovery, data sensing, public expression, risk warning, spatial analysis, and issue entry. It answers the question of how the city sees itself.

Manifestational architecture must avoid a single technological entrance. The city needs sensors, but it also needs residents. It needs platforms, but also street observation. It needs indicators, but also narratives. Only multiple modes of manifestation existing together can reduce blind spots.

Bearing Architecture

Bearing architecture includes spatial bearing, facility bearing, service bearing, institutional bearing, fiscal bearing, data bearing, ethical bearing, and community bearing. It answers the question of how the city responds to problems after they have been seen.

A good top-level design should clarify the bearing pathway for each type of problem. Do traffic problems enter road-network and public-transport optimization? Do issues of elderly people enter community services and age-friendly renovation? Do risk problems

enter infrastructure renewal and emergency plans? Do data problems enter governance institutions and security audits?

Reorganization Architecture

Reorganization architecture includes dynamic evaluation, modular replacement, institutional revision, algorithmic audit, public feedback, urban renewal, and operational iteration. It answers the question of how the city avoids its own rigidity.

The smart city should not be designed as a one-time project, but as a renewable system. The more grand a top-level design is, the more it must preserve channels of correction. The more unified a platform is, the more it must preserve local experimentation. The clearer standards become, the more they must allow new differences to enter.

Chapter Summary

This chapter expands top-level smart city design from technical architecture into manifestational architecture, bearing architecture, and reorganization architecture. Top-level design should first answer how the city sees and responds to difference, and only then how systems are built.

This shift prevents top-level design from becoming a list of platforms and turns it into a structural design for continuous urban learning.

Implications for Planning and Construction

- Top-level design documents should include a dedicated chapter on unmanifest difference analysis.
- In addition to technical architecture diagrams, manifestational pathways, bearing pathways, and reorganization pathways should be drawn.
- Top-level design should reserve mechanisms of iteration, audit, and exit.

Chapter 22

Digital Twins: Not Copying the City, but Manifesting the City

Misunderstandings of Digital Twins

Digital twin cities are often understood as three-dimensional models, urban sand tables, and visualization screens. Such presentation can help people understand urban form, but it can also reduce the digital twin to an exhibition project. Copying the appearance of the city is not the same as understanding the structure of the city.

In the Luminous City, the core of the digital twin is not whether it looks like the city, but what relations it manifests. It should help the city see dynamic relations among transportation, energy, ecology, population, services, risks, and institutions.

Manifesting Relations

A truly valuable digital twin should show the impacts of a planning decision on multiple systems: how a new commercial node changes traffic flow and noise; how public-service facilities affect elderly accessibility; how rainstorms spread along terrain and pipe networks; how high temperatures combine with aging and insufficient green space to create health risks.

Such relational manifestation allows the city to handle not only single-point events, but also causal chains, feedback loops, and level effects. The digital twin therefore becomes a vessel of manifestation, not merely a visual vessel.

Simulating Admissible Boundaries

One important value of the digital twin is to simulate admissible boundaries. It can help the city judge whether roads are approaching saturation, drainage is under pressure, electricity has peak risks, public services are imbalanced, or a policy will transfer pressure to other areas.

Through simulation, the city can see potential exhaustion before real damage occurs. But simulation is not decision itself. Model results must be combined with public discussion, professional judgment, and ethical boundaries before they become grounds for planning.

The Boundaries of Digital Twins

A digital twin cannot take copying everything as its goal. Many aspects of private life in the city should not be continuously copied and predicted. If the digital twin pursues the full mapping of persons, it violates the right to the unmanifest.

Digital twins should therefore be bounded by public problems, guided by necessary data, required to use explainable models, and grounded in public trust. They should manifest urban structures rather than turn every person into a transparent object.

Chapter Summary

This chapter redefines the digital twin: its core is not copying the appearance of the city, but manifesting structural relations and admissible boundaries of the city. Digital twins should help the city understand chains of influence and feedback loops.

At the same time, digital twins must accept ethical boundaries. They should manifest public problems rather than render private life fully transparent.

Implications for Planning and Construction

- Digital twin construction should prioritize the simulation of relations among transportation, ecology, services, and risks.
- Model outputs should be accompanied by explanatory notes and indications of uncertainty.
- Functions involving the mapping of individual behavior should undergo privacy impact assessment.

Chapter 23

Urban Operations: From Control Center to Regulation Center

The Positioning of the Operation Center

The urban operation center is an important facility of the smart city, but it should not be understood as an urban control room. If the operation center aims at total control, it treats the city as a machine that can be uniformly commanded. If it aims at regulation, it can become a coordinating hub for urban differences.

The task of the regulation center is to maintain admissible states among multiple goals: efficiency and fairness, speed and safety, manifestation and privacy, centralization and dispersion, normality and emergency, system optimization and individual dignity.

Traffic Regulation

Traffic operation is not simply the elimination of congestion. Congestion is often a composite manifestation of urban structure, jobs-housing relations, insufficient public transport, unfair road allocation, and an excessive radius of daily life. If signal optimization only increases vehicle speed, it may conceal deeper problems.

Traffic regulation in the Luminous City must combine vehicle efficiency, bus priority, slow-traffic safety, walking comfort, spatial mix, and demand management. Sometimes reducing unnecessary movement is wiser than accelerating movement.

Energy and Service Regulation

Energy operation is not unlimited supply, but the establishment of balance among load management, energy-saving renovation, distributed energy, low-carbon technology, and residents' behavior. The goal of smart energy is to keep urban operation from breaching ecological and fiscal admissibility.

Public-service operation is likewise not equal distribution. Different areas and groups have different needs. The smart city should form differentiated bearing according to population structure, degree of aging, distribution of children, changes in employment, and accessibility of services.

Risk Regulation

Emergency operation should not respond only after disaster occurs. It should manifest risks in advance, organize resources, protect vulnerable groups, and combine data warnings, spatial plans, departmental linkage, community mobilization, and public communication.

In the Luminous City, risk is not merely an abnormality to be suppressed, but a signal that an urban admissible boundary has been touched. Each risk-handling process should become a renewed understanding of the urban vessel.

Chapter Summary

This chapter repositions the urban operation center as a regulation center. It does not exist in order to dominate the city, but to maintain admissible states among multiple goals.

Traffic, energy, public-service, and risk operation should all move from single efficiency toward comprehensive bearing. The value of operation lies in regulating difference rather than flattening it.

Implications for Planning and Construction

- Operation-center indicators should display efficiency, bearing, risk, and fairness at the same time.
- After event handling, structural causes should be analyzed, rather than pursuing only immediate case closure.
- Urban operations should conduct regular cross-departmental reviews and transform operational data into planning adjustments.

Chapter 24

Urban Renewal: Repairing Exhausted Vessels

Renewal Is Not Decoration

Urban renewal is not the decoration of old space into new space, nor the replacement of life repair with commercial development. True renewal repairs urban vessels that can no longer bear differences of life. Aging buildings, insufficient facilities, lack of public space, service fractures, weakened relations, and ecological pressure all need to enter the logic of renewal.

The Luminous City treats renewal as reorganization, not replacement. Reorganization means preserving parts that can still bear, repairing parts that have become exhausted, introducing new bearing structures, and preventing new vessels from oppressing existing life again.

Renewal of Old Residential Compounds

The problems of old residential compounds are not only outdated facades, but insufficient bearing of life. Difficulty going upstairs and downstairs, aging pipelines, parking conflicts, inadequate public space, weak community care, fire hazards, and changing neighborly relations together constitute the decline of radiance in old residential compounds.

Smart renovation can provide access control, security, energy monitoring, and service platforms, but more important are age-friendly facilities, public space, grassroots services, resident deliberation, and long-term maintenance. Smart renovation without life repair only increases devices, not bearing.

Renewal of Industrial Spaces and Historic Districts

Renewal of industrial space cannot focus only on investment attraction and conceptual packaging. It must reorganize the relations among employment, transportation, housing, ecology, and public services. If industrial upgrading leads to jobs-housing separation, longer commutes, and displacement of existing groups, renewal produces new exhaustion.

Renewal of historic districts requires particular caution. History is not a consumable landscape but a vessel of urban memory. Commercial activation can bring vitality, but it

may also devour the lives of original residents. The Luminous City requires historic districts both to manifest memory and to bear present life.

Radiant Principles of Renewal

Urban renewal should follow three principles: allow concealed life to manifest again, allow exhausted space to bear again, and allow severed relations to reconnect. Any renewal that only raises land value, traffic, and visual image without improving the bearing of life is not true luminous renewal.

The success or failure of renewal should be judged by who remains afterward, who leaves, who gains convenience, who loses place, who can continue to participate, and who is turned into scenery. The Luminous City refuses to exchange the exhaustion of others' lives for superficial prosperity.

Chapter Summary

This chapter understands urban renewal as the repair of exhausted vessels rather than simple decoration or commercial replacement. Renewal should allow concealed life to manifest again and exhausted space to bear again.

Renewal of old residential compounds, industrial spaces, and historic districts should balance life-bearing, local memory, and publicness.

Implications for Planning and Construction

- Before urban renewal, original residents, service networks, and local memory should be assessed.
- Smart renovation should be designed together with age-friendliness, accessibility, public space, and long-term maintenance.
- Renewal effects should be judged by improvement of life, not only by investment, foot traffic, and landscape.

Chapter 25

Evaluation Systems: How to Judge Whether a City Has Radiance

Evaluation Cannot Focus Only on Technological Maturity

Smart city evaluation often focuses on platforms, data, applications, devices, standards, and operational efficiency. These dimensions are important, but they are insufficient for judging whether a city has radiance. Technological maturity may not synchronize with life improvement; data access may be disconnected from problem solving; the number of scenarios may have nothing to do with bearing capacity.

Evaluation of the Luminous City must place technical indicators back within an ontological framework. Technology is not an end in itself, but a tool of manifestation and bearing. Evaluation should ask whether technology helps the city see real differences, whether it enters institutional response, and whether it avoids new concealment and exhaustion.

Five Core Dimensions

This book proposes five dimensions of evaluation: degree of manifestation, degree of bearing, degree of admissibility, degree of ethics, and degree of reorganization. Degree of manifestation asks whether real problems are seen. Degree of bearing asks whether seen problems have structural responses. Degree of admissibility asks whether systems operate within thresholds. Degree of ethics asks whether the vulnerable, privacy, freedom, and dignity are protected. Degree of reorganization asks whether the city has the capacity for continuous correction.

These five dimensions do not replace existing standards, but provide theoretical calibration for them. They remind evaluators that the city is smart not because it is quantifiable, but because it can bear, renew, and respect.

Anti-Indicator Consciousness

All indicators simplify reality. The Luminous City does not reject indicators, but requires them to accept reflection. Every indicator should state its boundaries: what it measures, what it does not measure; what behavior it encourages, what behavior it may distort; and whether it causes selective construction or formalism.

Evaluation systems should retain qualitative investigation, case review, field observation, public feedback, and independent assessment. Without these supplements, indicators easily become new vessels of concealment.

Final Judgment

Whether a city has radiance ultimately depends not on how advanced its systems are, but on whether life is more capable of being borne, whether relations are more sustainable, whether difference has more dignity, whether risk is seen earlier, and whether exhaustion can be repaired.

The ultimate question of evaluation is this: does the city allow more lives to gain visibility, accessibility, safety, participation rights, and future possibility? If the answer is no, every title of intelligence needs to be re-examined.

Chapter Summary

This chapter proposes five evaluation dimensions for the Luminous City: manifestation, bearing, admissibility, ethics, and reorganization. They do not replace existing standards, but calibrate them.

Smart city evaluation must prevent technological maturity from replacing life improvement. The final question of evaluation is whether the city allows more lives to gain visibility, accessibility, safety, participation rights, and future possibility.

Implications for Planning and Construction

- Evaluation systems should put whether problems truly improve before whether systems have gone online.
- Every indicator should state its limitations and be accompanied by qualitative review.
- Indicators should be established for residents' feelings, accessibility for vulnerable groups, and long-term operational capacity.

Conclusion

The City Is Not an Endpoint, but Continuous Manifestation

Against the Fantasy of the Perfect City

There is no smart city completed once and for all. Every platform becomes outdated, every model drifts, every institution develops blind spots, every space ages, and every indicator is strategically adapted to. Understanding the smart city as a completed project misunderstands the living nature of the city.

The Luminous City opposes the fantasy of the perfect city. It does not promise to establish a system that never errs, knows everything, and is always efficient. It establishes an open structure that can continuously discover problems, bear differences, correct itself, and avoid exhaustion.

The City as an Open Living Structure

The city is an open living structure. It is composed of human bodies, space, technology, institutions, ecology, capital, culture, and memory. It is not simply a metaphorical organism, nor merely the opposite of a machine. It is a complex vessel in which difference continually enters manifestation, bearing, and reorganization.

In this sense, the wisdom of the city is not the wisdom of a central brain, but the wisdom of the whole structure. It is not the wisdom of a single algorithmic point, but the wisdom of multiple vessels in coordination. It is not the wisdom of controlling everything, but the wisdom of allowing life to continue generating.

The Direction of the Future City

The future city should not pursue absolute transparency, absolute efficiency, or absolute control. It should pursue deeper manifestation, more flexible bearing, clearer boundaries, and more continuous reorganization. Smart technology remains important, but it must be organized within human dignity, publicness, and ecological sustainability.

The future Luminous City should be able to let weak voices be heard, allow low-data existence to obtain services, reopen public space, respect privacy boundaries, manifest urban risks in advance, prevent renewal from devouring memory, allow institutions to learn from problems, and make technology a vessel of community.

Final Thesis

This book finally proposes that the Luminous City is not a city illuminated by technology, but a city in which differences of life continue to manifest within the vessels of space, institution, technology, and community while avoiding exhaustion.

The true intelligence of the smart city does not lie in how much data it can calculate, but in whether it can let real differences of life in the city be seen, borne, protected, and continuously renewed without exhausting itself. The wisdom of the city is not to control everything, but to allow complex life to continue shining within admissible structures.

Chapter Summary

This chapter completes the book's conclusion: the smart city is not an endpoint project, but an open process of continuous manifestation, bearing, constraint, and reorganization. Every platform, model, institution, and space will age, and therefore the city must retain its capacity for self-renewal.

The final thesis of the Luminous City is that urban wisdom does not lie in controlling everything, but in allowing complex life to continue shining within admissible structures.

Implications for Planning and Construction

- Smart city construction should move from project-based construction toward continuous operation and continuous learning.
- Urban planning should reserve space for future differences to enter and avoid excessive closure.
- All systems should undergo periodic review, correction, and reorganization.

Appendix I. Core Concepts of the Luminous City

This table explains the translation relations among the key terms of this book. It does not simply instrumentalize philosophical concepts, but shows how the same logic of manifestation obtains concrete forms in urban planning, construction, operation, and governance.

Concept	Meaning in the Ontology of the Luminous Will	Translation in the Smart City
Unmanifest	The precondition that has not yet entered manifestation	Unseen needs, risks, suffering, and potentials
Difference	The condition for the occurrence of manifestation	Heterogeneity among people, spaces, services, ecology, flows, and institutions
Will	The tendency by which difference continues without returning to silence	The persistence of urban needs, conflicts, memories, flows, and public claims
Radiance	Difference entering a visible, perceptible, and discussable state	Urban appearing in data, space, experience, and public issues
Vessel	The bearing structure that allows radiance to endure	Spaces, facilities, platforms, institutions, services, and communities
Admissibility	The boundary within which difference can sustainably unfold	Thresholds of transport, ecology, privacy, governance, finance, and social bearing
Exhaustion	The weakening that follows when a vessel loses bearing capacity	Difficult system maintenance, grassroots fatigue, spatial degradation, and declining trust
Reorganization	The repair, replacement, and regeneration of vessels	Urban renewal, institutional revision, algorithmic audit, and community co-governance

Appendix II. Ten Principles for Building a Luminous City

1. Do not begin from technology, but from unmanifest difference.
2. Do not equate data with reality.
3. Do not equate intelligence with control.
4. Do not equate platforms with the city.
5. Do not equate indicators with value.
6. Do not allow the vessel to overwhelm radiance.
7. Do not allow strong manifestation to become strong surveillance.
8. Do not allow efficiency to devour dignity.
9. Do not allow construction to replace operation.
10. Do not pursue a final city; pursue a city of continuous reorganization.

Appendix III. Planning and Construction Process for the Luminous City

Step	Name	Explanation
1	Identify the unmanifest	Through urban health checks, community interviews, field surveys, data analysis, public participation, and grassroots records, discover concealed needs, risks, suffering, and potentials.
2	Classify differences	Distinguish spatial, social, ecological, service, technological, governance, and digital-capacity differences, and judge which should be protected, regulated, or repaired.
3	Manifest problems	Use data platforms, maps, reports, public issues, digital twins, and community deliberation to bring problems into visible, sayable, and discussable states.
4	Establish bearing vessels	Translate problems into spatial projects, service systems, institutional procedures, technological tools, fiscal arrangements, and community mechanisms.
5	Set admissible boundaries	Clarify thresholds for transport, ecology, finance, governance, privacy, social endurance, and grassroots workload in order to prevent excessive consumption by systems.
6	Monitor exhaustion	Continuously observe technological, fiscal, governance, social, spatial, ecological, and meaning-based exhaustion.
7	Reorganize and renew	Through urban renewal, institutional revision, algorithmic audit, platform iteration, public feedback, and resilience renovation, restore the bearing capacity of urban vessels.

The above process can serve as an ontological calibration tool for smart city projects. It does not replace engineering implementation schemes; rather, it helps decision-makers prevent technology from preceding life, indicators from replacing value, construction from replacing operation, and platforms from overwhelming life.

Applied Use Example: Old Residential Compound Renewal

For an old residential compound renewal project, the process can be read concretely. The unmanifest differences may include stair-climbing difficulty, inaccessible routes, fire hazards, weak community care, inadequate public space, and residents who cannot use online services. Manifestation may be achieved through health checks, resident interviews, accessibility walks, maintenance records, and emergency-risk mapping.

The bearing vessels should then be specified: elevators or ramps for vertical mobility, shaded and seated public spaces for daily staying, community service points for care, fiscal arrangements for maintenance, offline service windows for residents with weak digital capacity, and deliberation procedures for conflicts over parking, access control, and renovation priorities. Admissibility review should ask whether privacy is protected, whether costs are sustainable, whether original residents can remain, and whether new devices increase or reduce governance burden.

This example shows how the Luminous City framework converts a smart project from equipment deployment into manifestational diagnosis, bearing design, boundary review, ethical correction, and long-term reorganization.

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Afterword

Luminous City attempts to bring the smart city back from technological engineering to urban existence itself. The city is worth planning, building, and governing because within it there is life, difference, vulnerability, memory, publicness, and a continually renewing future. Technology can help the city see itself, but it cannot decide the good for the city. True intelligence is that after seeing, the city can bear; after bearing, it can constrain; and after constraining, it can renew.